

Chapter 10: Gauge Theory:

Exercise #1: (10.16) "Gauge Transformation"
(pg 583)

$$F_{\mu\nu} \rightarrow F'_{\mu\nu} = g F_{\mu\nu} g^{-1}$$

(10.17) "Field Strength Tensor Transformation"

$$F \rightarrow F' \equiv g F g^{-1}$$

(10.13) "Field Strength Identity"

$$F = dA + A \wedge A$$

(10.12) "General Field Strength Tensor"

$$\begin{aligned} F_{\mu\nu} &= \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu] \\ F'_{\mu\nu} &= \partial_\mu A'_\nu - \partial_\nu A'_\mu + [A'_\mu, A'_\nu] \\ &= \partial_\mu (g A_\nu g^{-1} + g d g^{-1}) - \partial_\nu (g A_\mu g^{-1} + g d g^{-1}) + [g A_\mu g^{-1} + g d g^{-1}, g A_\nu g^{-1} + g d g^{-1}] \\ &= g (\partial_\mu A_\nu - \partial_\nu A_\mu) g^{-1} \\ &= g F_{\mu\nu} g^{-1} \end{aligned}$$

(pg 582)
"Abelian case"
 $[A_\mu, A_\nu] = 0$

Exercise #2: (10.5) "Transformation Identity"

(pg 584)

$$T'^{\mu\nu}_{\alpha\beta} = \frac{\partial y^\alpha}{\partial x^\mu} \frac{\partial x^\mu}{\partial y^\beta} \frac{\partial x^\nu}{\partial y^\beta} T^{\mu\nu}_{\alpha\beta} + \frac{\partial y^\alpha}{\partial x^\mu} \frac{\partial^2 x^\mu}{\partial y^\beta \partial y^\beta}$$

$$R_{\rho\sigma}^{\mu\nu} = \frac{\partial y^\alpha}{\partial x^\mu} \frac{\partial x^\nu}{\partial y^\rho} \frac{\partial x^\beta}{\partial y^\sigma} T_{\nu\rho}^{\mu\beta} + \frac{\partial y^\alpha}{\partial x^\rho} \frac{\partial^2 x^\rho}{\partial y^\sigma \partial y^\beta}$$

$$= \frac{\partial y^\alpha}{\partial x^\mu} \frac{\partial x^\nu}{\partial y^\rho} \frac{\partial x^\sigma}{\partial y^\beta} \frac{\partial y^\rho}{\partial y^\sigma} R_{\nu\rho}^{\mu\beta}$$

$$\text{when } \frac{\partial x^\sigma}{\partial y^\delta} \frac{\partial y^\rho}{\partial y^\delta} R_{\nu\rho}^{\mu\beta} = \frac{\partial x^\beta}{\partial y^\delta} T_{\nu\rho}^{\mu\beta} + \frac{\partial x^\mu}{\partial y^\alpha} \frac{\partial y^\beta}{\partial x^\nu} \frac{\partial y^\alpha}{\partial x^\rho} \frac{\partial^2 x^\rho}{\partial y^\delta \partial y^\beta}$$

Exercise #3: (Problem 1.8.2) "Total energy in Field"
(pg 606)

$$H = \dots = \frac{1}{2} \int d^3x (E^2 + B^2)$$

(10.42) "Ground state energy"

$$H_g = -J_e \sum_+ \hat{A}_g - J_m \sum_{\square} \hat{B}_\ell - J_s \sum_{n,l} \sigma_l(n,l)$$

The author related the two Hamiltonians above from 1.8.2 and 10.42.

With, $\hat{A}_g$ = Vector Potential

J_e = Current in Electric Potential

J_m = Separable current in
Magnetic field

B_ℓ = Magnetic Induction

$$= \underbrace{\frac{(\nabla \times B)B}{\mu_0}}_{\sim B^2} - \underbrace{J_s \sum \sigma_i(n_i \mu)}_{\text{"spin term"}}$$

Ampere's Law:

$$\nabla \times B = \mu_0 (J + \epsilon_0 \frac{\partial E}{\partial t})$$

(electro)statics
not (magneto)statics

Exercise #4: Perturbation:
(pg 607)

$$\nabla \times B = \mu_0 J$$

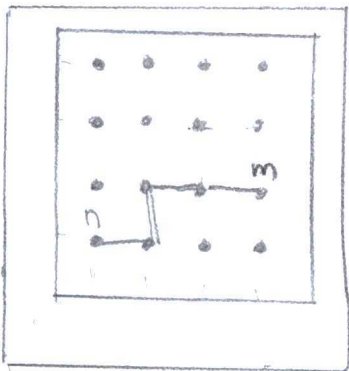
$$E = \langle n | H | n \rangle + \langle n | H' | n \rangle X + \frac{\langle n | H' | n \rangle^2}{E_n - E_m} X^2 + \dots$$

$$= \underbrace{E}_{0^{th} \text{ order}} + \underbrace{\frac{\partial}{\partial \mu} E \cdot X}_{1^{st} \text{ order}} + \underbrace{\frac{\partial^2}{\partial \mu^2} E \frac{X^2}{2!}}_{2^{nd} \text{ order}} + \dots$$

where $\langle n | H' | m \rangle = \text{"Magnetic current"}$
 $= J_m$

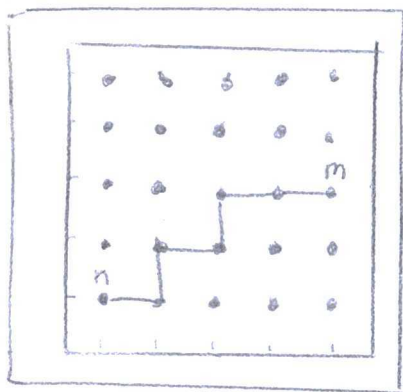
$E_n^{(0)} - E_m^{(0)} = \text{"Proportionality of strong energy"}$

$$= |n - m| 2 \cdot J_s$$



$$\frac{\langle n | H' | m \rangle^2}{E_n^{(0)} - E_m^{(0)}} = \frac{J_m^2}{14 - 11 \cdot 2 J_s}$$

$$= \frac{J_m^2}{6 J_s}$$



$$\frac{\langle n | H' | m \rangle^2}{E_0^{(0)} - E_R^{(0)}} = \frac{J_m^2}{13 - 11 \cdot 2 J_m}$$

$$= \frac{J_m^2}{4 \cdot J_s}$$

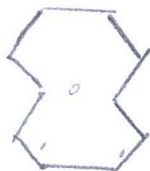
"Manhattan
metric distance"

Exercise #5:
(pg 609)

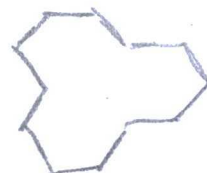
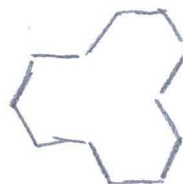
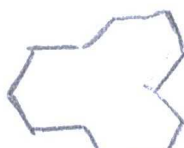
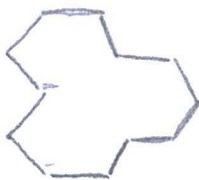
Ground
state



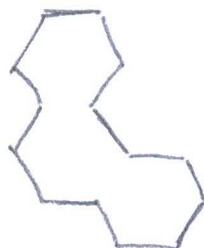
Second
state



Third
state

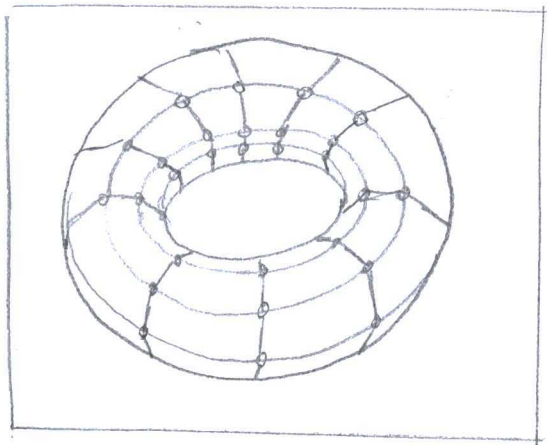


Fourth
state



...

Degrees of Freedom



"Torus - degrees of freedom"

① Vertices:

$$V = L \times L \quad \text{where } L = \text{vertices about } \theta/\phi$$

② Plaquette (unique vertices):

$$E = L \times L$$

③ Total Constraints:

$$(V-1) + (F-1) = (L^2-1) + (L^2-1) \\ = 2L^2 - 2$$

④ Degrees of Freedom:

$$2 \cdot L \times L = 2L^2$$

⑤ Degrees of Freedom

Remaining:

$$2L^2 - (2L^2 - 2) = 2$$

Problem 10.7.1 - Identifying the field strength tensor:

(from problem) "Loop"

$$x(t) = \{x^n(t)\}, \quad t \in [0, 1]$$

(10.11) "Transporter"

$$T[\gamma] = P \cdot e^{-\int_0^1 dt (\dot{x}^\mu \cdot A_\mu)}$$

Exponential

series:

$$e^x = 1 - x + \frac{x^2}{2!} - \dots$$

Expansion:

$$T[\gamma] = 1 - \int_0^1 dt \dot{x}^\mu(t) A_{\mu,t} + \frac{1}{2} \left(\int_0^1 dt \dot{x}^\mu(t) A_{\mu,t} \right)^2 -$$

$$= 1 - \int_{s(y)} dA + \frac{1}{2} \left(\int_{s(y)} dt \int_0^1 d\mu \dot{x}^\mu A_{\mu,t} \dot{x}^\nu A_{\mu,\nu} \right)$$

Problem 10.7.2 - Bianchi Identity:

a) (10.6) "Covariant Derivative"

$$D_\mu \equiv \partial_\mu + A_\mu$$

$$v^\mu \cdot D_\mu \cdot F_{\rho\sigma} = v^\mu (\partial_\mu + A_\mu) F_{\rho\sigma}$$

$$= v^\mu (\partial_\mu \cdot F_{\rho\sigma} + A_\mu \cdot F_{\rho\sigma})$$

$$D_\mu \cdot F_{\rho\sigma} = \partial_\mu \cdot F_{\rho\sigma} + A_\mu F_{\rho\sigma}$$

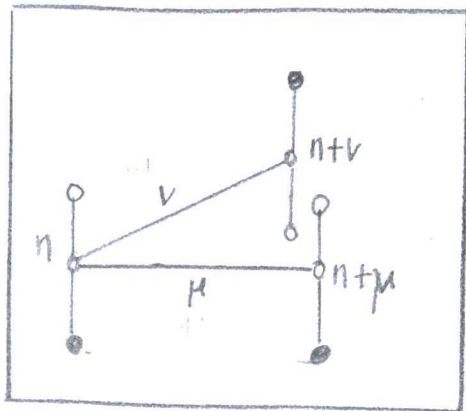
$$= dF + A \wedge F$$

$$b) D_\mu F_{\rho\sigma} + D_\rho F_{\sigma\mu} + D_\sigma F_{\mu\rho} =$$

$$= (2_\mu F_{\rho\sigma} + A_\mu F_{\rho\sigma}) + (2_\rho F_{\sigma\mu} + A_\rho F_{\sigma\mu}) + (2_\sigma F_{\mu\rho} + A_\sigma F_{\mu\rho})$$

$$= 0$$

Problem 10.7.3- Two-dimensional $\mathbb{Z}_2$ gauge Theory:



"Parallel
Transport"

(10.38) " $\mathbb{Z}_2$ -variant of the General action"

$$S[\sigma_3] = -J \sum_{n, \mu, \nu} \sigma_3(n, \mu) \sigma_3(n + \mu, \nu) \sigma_3(n + \mu + \nu, \mu) \sigma_3(n + \nu, -\nu)$$

while, $J = \epsilon\beta$

A reversal of an individual spin in the general action multiplies 10.38 by two because the pairs $\sigma_3(n, \mu)$ and $\sigma_3(n + K\nu, \mu)$. A coefficient derives from a reversal $Z(Z \circ J) = 4\epsilon\beta$.

The argument fails at $d=2$ and $d>2$ across a grid or links in a plaquette. The number of lines $[(L_\mu - 1) \times (L_\nu - 1)]$ in a given direction never exceeds the degrees of freedom $L_\nu(L_\mu - 1) + L_\mu(L_\nu - 1)$. The

10.7.4: Geometric Curvature from Gauge Theory:

Christoffel Symbols:

(10.32) "Metric for transport on a sphere"

$$T = \begin{pmatrix} T^\theta_\theta & T^\theta_\phi \\ T^\phi_\theta & T^\phi_\phi \end{pmatrix}$$

$$\text{When } T^\theta_\theta = 0$$

$$dT^\theta = 0$$

$$T^\theta_\phi = -\sin\theta\cos\theta d\phi \quad dT^\theta_\phi = (\sin^2\theta - \cos^2\theta)d\theta d\phi$$

$$T^\phi_\theta = \cot\theta d\phi$$

$$dT^\phi_\theta = -\csc\theta d\theta d\phi$$

$$T^\phi_\phi = 0$$

$$dT^\phi_\phi = 0$$

(10.14) "Curvature Tensor"

$$R = dT + T \wedge T$$

$$R_{ij} = dT^i_j + T^i_k \wedge T^k_j$$

$$\text{When } T^\theta_k \wedge T^k_\theta = 0$$

$$T^\theta_k \wedge T^k_\phi = \cos^2\theta d\theta \wedge d\phi$$

$$T^\phi_k \wedge T^k_\phi = \cot^2\theta d\theta \wedge d\phi$$

$$T^\phi_k \wedge T^k_\theta = 0$$

(From problem) "Riemann Curvature Tensor"

$$R = \begin{pmatrix} R^\theta_\theta & R^\theta_\phi \\ R^\phi_\theta & R^\phi_\phi \end{pmatrix}$$

$$= \begin{pmatrix} d\Gamma^\theta_\theta + \Gamma^\theta_\theta \wedge \Gamma^\theta_\theta & d\Gamma^\theta_\phi + \Gamma^\theta_\phi \wedge \Gamma^\theta_\phi \\ d\Gamma^\phi_\theta + \Gamma^\phi_\theta \wedge \Gamma^\phi_\theta & d\Gamma^\phi_\phi + \Gamma^\phi_\phi \wedge \Gamma^\phi_\phi \end{pmatrix}$$

$$= \begin{pmatrix} 0 & \sin^2\theta \\ -1 & 0 \end{pmatrix} d\theta \wedge d\phi$$

Coefficient to a Ricci Tensor:

(from problem) "Ricci Tensor"

$$R = \begin{pmatrix} R^\theta_\theta & R^\theta_\phi \\ R^\phi_\theta & R^\phi_\phi \end{pmatrix}, \text{ such that } R^\theta_\theta = 1$$

$$R^\theta_\phi = 0$$

$$R^\phi_\theta = 0$$

$$R^\phi_\phi = \sin^2\theta$$

$$\text{Diag}(R) = (1, \sin^2\theta) = g$$

Problem 10.7.5- Scalar Field Coupled to Gravity:

(10.53) "Minimal Two-derivative action
Form-invariant"

$$S_m[\phi, g] = -\frac{1}{2} \int d^4x \sqrt{-g} \partial_\mu \phi \partial^\mu \phi$$

where $g = \text{"Metric"}$

$\phi = \text{Function of field}$

Time-energy minimum Tensor:

$$\frac{\partial S[\phi, g]}{\partial g^{\mu\nu}} = -\frac{1}{2} \left(\frac{-1}{2} \right) \int d^4x (-g)^{-1/2} \partial_\mu \phi \partial^\mu \phi \frac{\partial g^{\mu\nu}}{\partial g^{\mu\nu}}$$

$$T^{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\partial S[\phi]}{\partial g^{\mu\nu}} = \frac{-2}{\sqrt{-g}} \left(\frac{1}{2} \right) \left(\frac{1}{2} \right) \int d^4x (\sqrt{-g})^{-1/2} \partial_\mu \phi \partial^\mu \phi$$

$$= \frac{1}{2\sqrt{-g}} \int d^4x \frac{1}{\sqrt{-g}} \partial_\mu \phi \partial^\mu \phi$$

$$= \partial_\mu \phi \partial_\nu \phi + \frac{1}{2\sqrt{-g}} \int 2\sqrt{-g} d^p \phi d_p \phi$$

$$= \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} g_{\mu\nu} \partial^p \phi \partial_p \phi$$

Unusual Integration by
Parts shown above:

$$dv = \frac{1}{\sqrt{-g}}$$

$$u = \partial_\mu \phi \partial^\mu \phi$$

$$v = -2\sqrt{-g}$$

$$du = \partial^p \phi \partial_p \phi$$

Problem 10.7.6 - Chern-Weil Theorem:

a) (from problem) "Monomials"

$$\text{In}(F)^n = \text{tr}(F^n)$$

$$\frac{dI_n(F^n)}{dF} = n \cdot \text{tr}(F^{n-1})$$

$$dI_n(F^n) = n \cdot \text{tr}(dF \wedge F^{n-1})$$

(10.52) "Bianchi Identity"

$$DF = dF + A \wedge F - F \wedge A$$

$$\begin{aligned} \text{tr}[A, F^n] &= \text{tr}(A \wedge F^n - F^n \wedge A) \\ &= n \text{tr}((A \wedge F - F \wedge A) F^{n-1}) \end{aligned}$$

$$= 0 \quad \text{...when commuting.}$$

$$\begin{aligned} \text{tr}[A, F^n] + dI_n(F^n) &= n \text{tr}((dF + A \wedge F - F \wedge A) F^{n-1}) \\ &= n \text{tr}(DF \cdot F^{n-1}) \\ &= 0 \end{aligned}$$

b) (10.13) "Field Strength"

$$F = dA + A \wedge A$$

$$\frac{dI(F_t)}{dt} = \frac{d}{dk_t} \text{ Relationships}$$

$$\begin{aligned} \frac{dI(F_t)}{dt} &= n I(\overset{\circ}{F} \cdot \overset{\circ}{F}^{n-1}) \\ &= n \text{tr}((d\overset{\circ}{A} + \overset{\circ}{A} \wedge \overset{\circ}{A} + \overset{\circ}{A} \wedge \overset{\circ}{A}) \wedge \overset{\circ}{F}^{n-1}) \\ &= n \text{tr}(d\overset{\circ}{A} \wedge \overset{\circ}{F}^{n-1} + \overset{\circ}{A} \wedge (A \wedge F^{n-1} - F^{n-1} \wedge A)) \end{aligned}$$

$$\begin{aligned}
&= n \operatorname{tr} (d\dot{A} \wedge F^{n-1} + \dot{A} \wedge \left(\sum_{\ell=0}^{n-2} F^\ell \wedge (A \wedge F - F \wedge A) \wedge F^{n-2-\ell} \right)) \\
&= n \operatorname{tr} (d\dot{A} \wedge F^{n-1} + \dot{A} \wedge \left(- \sum_{\ell=0}^{n-1} F^\ell \wedge (dF) \wedge F^{n-2-\ell} \right)) \\
&= n \operatorname{tr} (d\dot{A} \wedge F^{n-1} + \dot{A} \wedge -dF^{n-1}) \\
&= n \cdot d \operatorname{tr} (\dot{A} \wedge F^{n-1}) \\
&= d_{K_E} \quad \leftarrow \quad d_{K_E} = d/dK_E \text{ much like } \partial_t = \partial/\partial t
\end{aligned}$$

Exactness of $I_n(F) - I_n(F')$:

$$I_n(F) - I_n(F') = d \int_0^1 dt K_t \quad \text{is an exact}$$

integral between the interior and exterior derivative.

If $n=1$, then

$$\begin{aligned}
I(F) - I(F') &= d \int_0^1 dt K_t \\
&= d \int_0^1 dt n \operatorname{tr} (A \wedge (t dA + t^2 A \wedge A))^{n-1} \\
&= n \cdot d \operatorname{tr} (A)
\end{aligned}$$

If $n=2$, then

$$I(F) - I(F') = d \int_0^1 dt K_t$$

$$= d \int_0^1 dt \operatorname{tr} (A^\wedge (t dA + t^2 A^\wedge A)^{2-1})$$

$$= d \int_0^1 dt \operatorname{tr} (A^\wedge (t dA + t^2 A^\wedge A))$$

$$= \frac{\hbar}{2} d \operatorname{tr} (A^\wedge dA + \frac{2}{3} A^\wedge A^\wedge A)$$

Both $n=1$ and $n=2$ are proportional to Chern numbers in equation 10.49.

(10.49) "Chern numbers"

$$Ch_1(F) = \frac{1}{2\pi} \operatorname{tr}(F) = \frac{i}{2\pi} d \operatorname{tr} A$$

$$Ch_2(F) = -\frac{1}{8\pi^2} \operatorname{tr}(F^\wedge F)$$

$$= -\frac{1}{8\pi^2} d \operatorname{tr} (A^\wedge dA + \frac{2}{3} A^\wedge A^\wedge A)$$

Problem 10.7.7: Gauge invariant of Chern-Simons Action

(10.27) "Chern-Simons Action"

$$S_{CS}^{(1)} = i k \int_M \operatorname{tr}(A)$$

$$S_{CS}^{(3)} = \frac{k}{4\pi} \int_M \operatorname{tr} (A^\wedge dA + \frac{2}{3} A^\wedge A^\wedge A)$$

(10.50) "Chern number"

$$Ch_n[F] = \int_M Ch_n(F) \quad \text{where } F = \begin{matrix} \text{"Field Strength"} \\ \text{Tensor"} \end{matrix}$$

(10.48) "Chern Character"

$$\text{Chn}(F) = \frac{1}{n!} \text{tr} \left(\frac{iF}{2\pi} \right)^n$$

(10.3) "Gauge Transformation"

$$A' = g A g^{-1} + g dg^{-1}$$

a) If $g^{-1}dg = \phi$, then

$$\text{tr}(A' \wedge A' \wedge A' - A \wedge A \wedge A)$$

$$= \text{tr}((g A g^{-1} + g dg^{-1}) \wedge (g A g^{-1} + g dg^{-1})$$

$$\wedge (g A g^{-1} + g dg^{-1}) - A \wedge A \wedge A)$$

$$= \text{tr}((A + \phi) \wedge (A + \phi) \wedge (A + \phi) - A \wedge A \wedge A)$$

$$= (3A \wedge \phi \wedge \phi + 3A \wedge A \wedge \phi + \phi \wedge \phi \wedge \phi)$$

$$b) \text{tr}(A' \wedge dA' - A \wedge dA) = \text{tr}((A + \phi) \wedge d(A + \phi) - A \wedge dA)$$

$$= \text{tr}((A + \phi) \wedge (-\phi \wedge \phi) \wedge A \\ + \phi \wedge dA - A \wedge dA)$$

"Whew, tired."